رواد مصر الرقمية

DIGITAL EGYPT PIONEERS INITIATIVE (DEPI)

DEPI | PROFESSIONAL DATA ANALYTICS PROJECT

The Relationship Between Obesity and Health, Lifestyle, and Economic Factors

A Comparative Data Analysis of Egypt, Japan, and the United States

Submitted as part of: DEPI Professional Data Analytics Track

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PROJECT PLANNING & MANAGEMENT

1. Project Overview

This project explores the relationship between adult obesity prevalence and selected health, lifestyle, and economic indicators in Egypt, Japan, and the United States from 2000 to 2022. Publicly available secondary data will be analyzed to compare trends and explore associations, subject to data availability and comparability.

2. Problem Statement

Adult obesity rates vary across countries and over time. The project will examine whether changes in obesity prevalence occur alongside differences in economic conditions, healthcare spending, diabetes prevalence, and physical inactivity. The analysis is descriptive and will not claim that an observed association proves causation.

3. Aim and Objectives

Aim: Compare obesity trends across Egypt, Japan, and the United States and explore their relationships with selected health, lifestyle, and economic indicators.

- Collect comparable country-level data for 2000–2022 where available.
- Check data quality, definitions, units, missing values, and year coverage.
- Use SQL to organize and query the dataset.
- Use Python and Pandas for cleaning, descriptive statistics, and exploratory analysis.
- Use Tableau for focused exploratory visualizations and Power BI for the final interactive dashboard.
- Communicate findings and limitations without unsupported causal conclusions.

4. Project Scope

Included:

- Countries: Egypt, Japan, and the United States; period: 2000–2022, subject to data availability.
- Main outcome: age-standardized adult obesity prevalence (BMI ≥ 30 kg/m²), using a harmonized series where possible.
- Five core indicators proposed for the updated scope: adult obesity prevalence; GDP per capita (PPP); diabetes prevalence; insufficient physical activity among adults; and health expenditure per capita. These indicators remain provisional until coverage, definitions, and comparability are checked.
- Tools: SQL, Python (Pandas/NumPy and visualization libraries), Tableau, Power BI, and GitHub.

Excluded from the current scope:

- Estimating total economic costs attributable to obesity, productivity losses, or employment effects.
- Machine-learning prediction and complex causal modelling.
- Claims that an economic or lifestyle indicator directly causes obesity.

5. Methodology and Main Activities

- **Planning:** confirm scope, responsibilities, schedule, and deliverables.
- **Data collection:** download data and document each source, definition, unit, and coverage.
- **Data preparation:** check data types, missing values, duplicates, country names, years, and comparability.
- **Data integration:** combine indicators into a country-year dataset and load/query data using SQL.
- **Analysis:** calculate descriptive statistics, compare trends, and explore associations in Python.
- **Visualization:** create exploratory charts in Tableau and a final interactive Power BI dashboard.
- **Validation and reporting:** verify outputs, document limitations, prepare the report and presentation, and publish project files on GitHub.

6. Proposed Work Plan / Gantt Schedule

This is a proposed six-week sequence for the whole team. All six members participate in every phase; the schedule shows the main project activities over time, not separate assignments for individual members. The agreed deadline remains unchanged.

Activity	W1	W2	W3	W4	W5	W6
Planning and requirements	●					
Literature review	●	●				
Data collection and source log	●	●				
Data cleaning and SQL integration		●	●			
Exploratory analysis			●	●		
Tableau visualizations				●	●	
Power BI dashboard				●	●	●
Testing, report, presentation, GitHub					●	●

7. Team Roles and Responsibilities

All six team members will work collaboratively throughout the entire project lifecycle, including planning, literature review, requirements gathering, data collection, data cleaning, analysis, visualization, testing, and final presentation. The team will discuss key decisions together, share knowledge, review each other's work, and ensure that all members understand the project's objectives, methodology, tools, and findings.

To maintain organization and meet deadlines, coordination tasks such as tracking progress, organizing shared files, and collecting meeting notes may be assigned to individual members. These are coordination duties only and do not restrict participation in any project phase. All major deliverables and decisions will be reviewed and agreed upon by the team.

Coordination activity	Team approach
Progress tracking	One member may update the shared task tracker; all members review it.
File organization	Keep source data, scripts, reports, and dashboard files in a shared folder/repository.
Peer review	Members review each other's work before merging it into the final deliverables.
Decision-making	Discuss and agree on scope, indicators, analysis choices, and final conclusions as a team.

8. Risk Assessment

Risk	Likelihood / Impact	Mitigation
Missing country-year data	Medium / High	Check coverage early; document gaps; do not invent or silently interpolate values.
Different definitions or units	Medium / High	Record source, age group, standardization, unit, and methodology; use comparable indicators.
Delays in data collection	Medium / Medium	Save raw files, maintain a shared source log, and set internal checkpoints.
Errors in cleaning or analysis	Medium / High	Keep raw data unchanged, use reproducible scripts, validate outputs, and arrange peer review.
Uneven workload or missed deadlines	Medium / Medium	Assign task owners, track progress at check-ins, and communicate blockers early.
Unclear or misleading dashboard	Low–Medium / Medium	Use clear labels, units, filters, source notes, and non-causal wording.

9. KPIs and Success Criteria

KPI	Success criterion
Deliverable completion	Agreed deliverables are submitted by confirmed deadlines.
Data coverage	All three countries are represented where data are available; gaps are documented.
Data quality	No unresolved duplicate country-year records in the final dataset; units and types are checked.
Traceability	Every indicator has a documented source, definition, unit, and coverage period.
Reproducibility	SQL queries and Python scripts are saved and can be rerun by another team member.
Dashboard usability	Charts have clear titles, labels, units, country/year comparisons, and source notes.
Team coordination	Every task has an owner and status; blockers are communicated regularly.

10. Limitations and Assumptions

- The analysis depends on the availability and comparability of public secondary data; the five core indicators are provisional until data checks are complete.
- Some indicators may not cover every year from 2000 to 2022; coverage will be reported for each indicator.
- Country-level patterns cannot establish individual-level relationships.
- Correlation and trend comparisons do not prove causation.
- The six-week schedule and role allocation are proposals to be confirmed by the team and instructor.

11. Planned Deliverables

- Project planning and management document; literature review; requirements gathering document.
- Cleaned dataset and source log; SQL files and Python analysis scripts.
- Tableau exploratory visualizations and Power BI dashboard.
- Testing notes, final report, presentation, and GitHub repository.

12. Starting Points for Data Sources

- Obesity prevalence — Our World in Data / NCD-RisC: https://ourworldindata.org/obesity
- GDP per capita, PPP — World Bank: https://data.worldbank.org/indicator/NY.GDP.PCAP.PP.CD
- Health expenditure per capita, PPP — World Bank / WHO GHED: https://data.worldbank.org/indicator/SH.XPD.CHEX.PP.CD
- Physical activity / insufficient physical activity — WHO: https://www.who.int/data/gho
- Population aged 65+ — World Bank: https://data.worldbank.org/indicator/SP.POP.65UP.TO
- Urban population share — World Bank: https://data.worldbank.org/indicator/SP.URB.TOTL.IN.ZS

LITERATURE REVIEW

Global Obesity Data Analysis using Python, MySQL & Power BI

Review of the GitHub project by ayushzxx

1. Selected Project and Review Approach

This review examines the GitHub project “Global Obesity Data Analysis using Python, MySQL & Power BI” by ayushzxx. It is based on the publicly available README description and linked repository information. The project was not independently executed or tested; therefore, the workflow described below reflects the published documentation, while the strengths and limitations are an evaluation of the documented approach rather than verification of its results.

2. Project Objective and Analytical Focus

According to its description, the project explores obesity data across countries and years and presents the results through a Power BI dashboard. The described dashboard supports comparisons by country, year, and gender. This is relevant as an example of turning a dataset into structured analysis and an accessible visual output. However, the available description does not establish that the project investigates economic or healthcare factors associated with obesity, so its focus should be understood primarily as obesity-trend analysis.

3. Data Workflow and Tools

The documented workflow links three stages: data preparation in Python, storage and querying in MySQL, and visual presentation in Power BI. Python can support inspection, cleaning, transformation, and preparation; a relational database can organize records and support structured queries; and Power BI can present comparisons through charts and dashboard filters. The README-based review does not confirm the exact cleaning rules, database schema, SQL query design, or validation procedures used.

For country-and-year analysis, a useful design principle is to document country, year, indicator, unit, population group, and source. If gender-specific results are used, gender categories and definitions should also be recorded. Before combining indicators, analysts should check that time coverage and definitions are compatible. These are recommendations for adapting the workflow, not claims that the reviewed project has implemented each practice.

4. Strengths of the Documented Approach

- **End-to-end workflow.** The description connects data preparation, database storage, and dashboard presentation, illustrating a workflow from raw data to user-facing output.
- **Country and time comparisons.** Comparing countries and years can reveal differences in levels and trends when definitions and data coverage are consistent.
- **Interactive communication.** The reported Power BI dashboard and country, year, and gender filters can make results easier to explore than static charts, provided labels, units, and sources are clear.

- **Transferable technical skills.** The described use of Python, MySQL, and Power BI provides a practical reference for combining programming, SQL-based data handling, and visualization.

5. Limitations and Questions for Further Verification

The main limitation is the available evidence: a README description is not enough to verify calculation accuracy, underlying data quality, or dashboard reliability. A stronger technical assessment would require inspecting the source dataset, code, SQL queries, transformations, and validation checks.

- **Data provenance and definitions.** Check source, collection method, population coverage, indicator definition, and measurement unit; differences can make comparisons misleading.
- **Missing values and coverage.** Available years may differ by country, gender, or indicator. Missing values should be documented rather than treated as zero or filled without justification.
- **Cleaning and transformation.** Cleaning steps should be reproducible, with clear handling of duplicates, invalid records, outliers, and special missing-value codes.
- **Calculation transparency.** Averages, percentages, and changes should have documented formulas and denominators; aggregated values should represent compatible populations and periods.
- **Dashboard interpretation.** A clear design does not prove analytical accuracy. Charts should identify indicators, units, year ranges, sources, and caveats.
- **Scope of inference.** Trend comparisons describe patterns but cannot establish why obesity changed. Associations should not be presented as proof of causation.

6. Comparison with Our Group Project

Both projects examine obesity-related data across countries and over time and communicate results through analysis and visualization. The reviewed project, as described in its README, focuses on obesity comparisons by country, year, and gender. Our group project will explore relationships between adult obesity prevalence and selected health, lifestyle, and economic indicators in Egypt, Japan, and the United States from 2000 to 2022, subject to data availability and comparability.

Our proposed core indicators are adult obesity prevalence, GDP per capita (PPP), diabetes prevalence, insufficient physical activity among adults, and health expenditure per capita. These remain provisional until source coverage, definitions, units, and comparability are checked. The project aims to describe trends and associations, not to claim that a single economic or lifestyle indicator causes obesity.

Our planned toolchain includes SQL for integration and querying; Python and Pandas for cleaning, calculations, and exploratory analysis; Tableau for focused exploratory visualizations and a data story; and Power BI for the final interactive dashboard. This assigns different purposes to Tableau and Power BI. The final implementation will depend on the agreed scope and available data.

7. Lessons and Actions for Our Project

1. Create a source log for every indicator, including URL, definition, unit, population group, available years, and download date.
2. Build a country-year dataset with a clear key; inspect duplicates, missing values, data types, and inconsistent country names before integration.
3. Keep raw source files unchanged and document cleaning and transformation steps so another team member can reproduce results.
4. Compare countries only after checking indicator definitions, age groups, units, and standardization methods.
5. Use clear, separately labelled visualizations for obesity trends and comparisons with economic or health indicators, with source notes.
6. Validate key calculations independently and have team members review SQL queries, Python outputs, and dashboard values.
7. Distinguish association from causation and explain data gaps or limitations that affect comparisons.

8. Overall Assessment

The reviewed project is a useful reference for preparing obesity data, storing it in a database, and presenting comparisons in a dashboard. Its strongest contribution is the workflow linking data preparation, SQL-based storage, and visualization. Its relevance is limited by the difference in research scope and by the fact that this review has not independently tested the implementation or outputs. Our project can adapt the workflow while adding selected health, lifestyle, and economic indicators, stronger data-quality documentation, and explicit limits on interpretation.

LITERATURE REVIEW

Time Series: The World Bank Global Health

Global Overweight and Undernourishment

Project reviewed: Ísis Santos Costa, “Time Series: The World Bank Global Health” / “Global Overweight & Undernourishment.” Published in 2024.

Purpose of this review: To examine the project’s approach, evaluate its strengths and weaknesses, suggest improvements, and outline grading criteria for documentation, implementation, testing, and presentation.

This review focuses on the notebook and its available code, explanations, and outputs. The grading table is a suggested rubric because no official mark distribution was provided.

1. Project Overview

The project uses public World Bank Global Health data to explore trends in overweight and undernourishment. It compares high-income and low-income groups and uses time-series charts to show how the indicators change over time. The author also applies a forecasting method to forecast selected indicators.

The notebook describes a dataset with hundreds of health indicators and records covering many countries and years. For this analysis, it selects indicators for adult overweight and undernourishment, then focuses on high-income and low-income groups. The main chart is labelled 1975–2019, although the number of available observations differs between indicators.

The tools used include Python, Pandas, SQL through Google BigQuery, Matplotlib, and statsmodels. The notebook does not demonstrate a MySQL database workflow or an interactive Power BI/Tableau dashboard.

2. Methodology

- Data selection: The author queries the World Bank Global Health dataset and selects indicators related to overweight and undernourishment.
- Data preparation: The selected records are filtered and arranged by year for analysis.
- Visualization: Line charts are used to compare changes over time between income groups.
- Forecasting: a forecasting method is applied to selected series to estimate future values.

The approach is useful for exploring long-term trends. However, the notebook’s data-cleaning choices and forecast accuracy need further checking before the results are treated as reliable.

3. Main Findings Reported in the Notebook

- The author reports that overweight prevalence increased over time in both high-income and low-income groups.
- The notebook reports a larger percentage increase in overweight in the low-income group over the period studied.
- The author observes a decline in undernourishment for part of the period, followed by an increase in more recent years.
- The forecasts suggest that the selected trends may continue over the forecast period.

These points summarize the author's interpretation. The forecasts are projections based on historical data and should not be treated as certain future outcomes.

4. Feedback & Evaluation

The project is a useful example of time-series health data analysis. It has a clear topic, uses a public dataset, and combines SQL queries, Python, and visualizations. The forecasting section adds another analytical method. However, some parts of the work need clearer explanation and stronger validation.

Area	Evaluation
Project objective	The opening question mentions factors affecting longevity, but the analysis mainly examines overweight and undernourishment. The objective and analysis should match more closely.
Data quality	The notebook removes records with a value of 1 because the author suspects these may represent missing data. The source coding should be checked first to avoid removing valid observations.
Data coverage	The selected indicators have different numbers of observations, so they may not cover the same years. This should be stated clearly.
Forecasting	The notebook estimates future values, but it should explain more clearly how close these estimates are to actual values.
Visualization	The line charts help show trends, but an interactive dashboard would make it easier to filter and compare indicators.
Relevance	The project compares income groups, not Egypt, Japan, and the United States, and it does not examine GDP per capita or healthcare spending in detail.

Overall, the notebook works well as an example of time-series analysis, but it would benefit from clearer data-cleaning rules, clearer explanation of future estimates, and a closer match between its research question and the indicators analyzed.

5. Suggested Improvements

- Clarify the objective: Focus the research question on overweight and undernourishment trends, or include indicators that directly measure the stated question about longevity.
- Check data cleaning: Confirm how missing values are represented in the original source before removing values equal to 1.
- Document the dataset: Add a short data dictionary with indicator names, units, definitions, source links, and the available years.
- Explain forecast results more clearly: Describe that future estimates may not match actual values.
- Compare individual countries: Include Egypt, Japan, and the United States if the aim is to support a country-level project.
- Add economic indicators: Study obesity alongside GDP per capita and healthcare expenditure where comparable data are available.
- Add the planned tools: Use MySQL for structured storage and SQL analysis, then build a Power BI or Tableau dashboard for interactive comparisons.

6. Final Grading Criteria

The following marking scheme is a suggested way to assess the project. It is not an official grade awarded to the author.

Category	Marks	What should be assessed
Documentation	25	Clear objective, data sources, indicator definitions, methodology, references, and limitations.
Implementation	30	Correct data extraction, cleaning, Python analysis, SQL queries, and suitable visualizations.
Testing	20	Data-quality checks, verification of missing values, reproducible code, and clear checking of the results.
Presentation	25	Readable charts, clear explanations, logical structure, and conclusions supported by the analysis.
Total	**100**	**Suggested allocation; use the lecturer's official distribution if one is provided.**

7. Relevance to Our Graduation Project

Our graduation project studies the relationship between adult obesity prevalence, healthcare indicators, and economic indicators in Egypt, Japan, and the United States. It focuses on trends, comparisons, and correlations rather than claiming that one factor directly causes another.

The World Bank notebook can help with selecting health indicators, organizing time-series data, writing SQL queries, and creating trend charts. However, our project needs a different country comparison and a wider set of indicators, including GDP per capita and healthcare expenditure. It also plans to use MySQL and Power BI, which are not demonstrated in this notebook.

The main lesson is to use a consistent data source where possible, document the meaning and coverage of each indicator, and avoid making conclusions that go beyond what the data supports.

8. Conclusion

The World Bank Global Health notebook is a useful reference for exploring changes in overweight and undernourishment over time. Its strengths are the use of public data, SQL-based selection, Python visualization, and an attempt to forecast trends. Its main weaknesses are the need for clearer data-cleaning decisions, stronger forecast evaluation, and closer alignment between the question and the analysis.

For our graduation project, its time-series approach can be adapted and combined with MySQL, additional economic and healthcare indicators, and a Power BI dashboard to meet our project objectives.

Requirements Gathering

1. Stakeholder Analysis

Stakeholder analysis identifies the individuals and groups involved in the project, their interests, and their requirements.

Stakeholder	Needs and Expectations
Project Team	Clear requirements, organized datasets, defined responsibilities, and effective collaboration throughout the project.
Project Supervisor	Accurate analysis, appropriate methodology, documented sources, and completed project deliverables.
Data Analysts	Reliable data, consistent indicator definitions, data-cleaning procedures, and analytical tools.
Public Health Researchers	Comparisons of obesity trends and their relationships with economic, lifestyle, healthcare, and demographic indicators.
Healthcare Policymakers	Evidence-based insights that may help identify public health trends and inform further investigation.
Dashboard Users	Clear visualizations, interactive filters, country comparisons, and understandable findings.

2. User Stories and Use Cases

2.1 User Stories

User stories describe the system's expected capabilities from the perspective of its users.

- **US-01:** As a data analyst, I want to collect data from reliable sources so that I can conduct an evidence-based analysis.
- **US-02:** As a data analyst, I want to clean and integrate datasets so that the data is consistent and ready for analysis.
- **US-03:** As a data analyst, I want to compare obesity prevalence in Egypt, Japan, and the United States so that I can identify differences in national trends.
- **US-04:** As a data analyst, I want to explore the relationship between obesity, GDP per capita, and healthcare expenditure so that I can identify potential associations.
- **US-05:** As a data analyst, I want to examine physical inactivity, dietary indicators, and demographic factors alongside obesity trends so that I can explore additional patterns.
- **US-06:** As a dashboard user, I want to filter results by country and year so that I can focus on specific comparisons.
- **US-07:** As a dashboard user, I want to view interactive charts and summary indicators so that I can understand the main findings easily.
- **US-08:** As a project supervisor, I want to review the sources, calculations, and documentation so that I can evaluate the accuracy and reliability of the project.

2.2 Use Cases

Use Case 1: Collect and Prepare Data

- **Actor:** Data Analyst
- **Objective:** Prepare a reliable dataset for analysis.
- **Preconditions:** Relevant public data sources have been identified.
- **Main Flow:**
    1. Collect data from selected sources.
    2. Record indicator definitions, units, and source URLs.
    3. Inspect missing values, duplicates, and inconsistent formats.
    4. Standardize compatible variables.
    5. Integrate datasets using country and year.
    6. Validate and save the cleaned dataset.
- **Postconditions:** A documented, analysis-ready dataset is available.

Use Case 2: Analyze Obesity Trends

- **Actor:** Data Analyst / Researcher
- **Objective:** Explore obesity trends and related indicators.
- **Preconditions:** The cleaned dataset is available.
- **Main Flow:**
    1. Load the dataset into Python or SQL.
    2. Calculate descriptive statistics and changes over time.
    3. Compare obesity prevalence across the three countries.
    4. Examine relationships with income, healthcare expenditure, and lifestyle indicators.
    5. Generate charts and document the findings.
- **Postconditions:** Analytical results and exploratory visualizations are produced.

Use Case 3: Explore the Interactive Dashboard

- **Actor:** Dashboard User
- **Objective:** Explore and compare the results visually.
- **Preconditions:** The Power BI dashboard has been developed and tested.
- **Main Flow:**
    1. Open the dashboard.
    2. View summary indicators and trend charts.
    3. Select a country or year range.
    4. Explore comparisons between obesity and other indicators.
    5. Review the displayed findings and data limitations.
- **Postconditions:** The user can explore the available results and understand the main patterns.

3. Functional Requirements

Functional requirements define what the proposed analytics solution must do.

ID	Requirement
FR-01	The system shall support the analysis of Egypt, Japan, and the United States from 2000 to 2022, subject to data availability.
FR-02	The team shall collect data from reliable public sources and document the sources.
FR-03	The data preparation workflow shall identify missing values, duplicates, and inconsistent data formats.
FR-04	The workflow shall standardize compatible variables and integrate datasets using country and year.
FR-05	The system shall provide an analysis-ready dataset containing the selected indicators.
FR-06	The analysis shall calculate descriptive statistics and selected changes in indicators over time.
FR-07	The analysis shall compare adult obesity prevalence across the three countries.
FR-08	The analysis shall explore associations between obesity prevalence and income, healthcare expenditure, physical inactivity, and available dietary indicators.
FR-09	The analysis shall consider demographic indicators, including population aged 65+ and urban population share, where data is available.
FR-10	SQL shall be used for data integration, querying, aggregation, and analytical summaries.
FR-11	Python shall be used for data preparation, calculations, exploratory analysis, and visualization.
FR-12	Tableau shall be used to create exploratory visualizations and a focused data story.
FR-13	Power BI shall provide an interactive dashboard with relevant charts and summary indicators.
FR-14	The dashboard shall allow users to filter results by country, year, and relevant indicators where applicable.
FR-15	The project shall document data sources, calculations, missing values, and analytical limitations.
FR-16	The team shall validate important calculations and maintain project documentation and code in GitHub.

4. Non-Functional Requirements

1. Performance

- **NFR-01:** The system should process the dataset efficiently.
- **NFR-02:** The dashboard should respond quickly when users apply filters.

2. Security

- **NFR-03:** The project should use publicly available data without personal information.
- **NFR-04:** Sensitive information, such as passwords, must not be shared on GitHub.

3. Usability

- **NFR-05:** Charts and dashboards should be clear and easy to understand.
- **NFR-06:** The dashboard should allow users to explore data easily.

4. Reliability and Accuracy

- **NFR-07:** Calculations and analysis results should be checked for accuracy.
- **NFR-08:** Data sources and project limitations should be documented clearly.

5. Maintainability

- **NFR-09:** Code and project files should be organized and documented.
- **NFR-10:** GitHub should be used to track project changes.

Conclusion

The requirements define the stakeholders, expected user interactions, system functionalities, and quality standards for the project. They provide a foundation for developing a reliable and reproducible data analytics solution that compares obesity trends and explores their relationships with economic, healthcare, lifestyle, and demographic indicators in Egypt, Japan, and the United States.

The analysis will depend on the availability and comparability of the selected data sources, and all findings will be interpreted without making unsupported causal claims.

References

[1] ayushzxx. Global Obesity Data Analysis using Python, MySQL & Power BI. GitHub repository, README. Accessed October 16, 2026.

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[3] Google BigQuery Public Dataset Marketplace. The World Bank: Global Health. https://console.cloud.google.com/marketplace/product/the-world-bank/global-health